INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
TEST SET, FUEL SYSTEM

1. SCOPE. This Commercial Item Description, (CID), describes an AC Fuel Quantity Test Set based on Bridge Capacitance/Resistance Test Set (BC/RFGTS) technology herein referred to as test set. This CID describes a minimum requirement for the test set in which only those manufacturers that meet or exceed the following requirements are qualified per this CID. The test set should be capable of measuring fuel tank resistance and capacitance, simulating tank unit capacitance, simulating the compensator capacitance for testing with dry fuel tanks, and be used as a standard to calibrate the fuel measurement systems and compensator capacitance.

2. SALIENT CHARACTERISTICS. The test set shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The test set shall be Class 1 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The test set shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.2.1 Intrinsic Safety. Current shall be limited to 10 milliamperes (ma) in all modes. Capacitive discharge energy shall be limited to 0.02 milli-joules at 0.016 µF for all modes of operation.

2.2.2 Explosive Atmosphere. The test set shall meet the explosive atmosphere requirements of MIL-PRF-28800, or the requirements of MIL-STD-810F Change Notice 3, METHOD 511.4 tailored for the same tests performed for MIL-PRF-28800.

2.2.3 Leakage Current Protection. Protection shall be provided from leakage current in excess of 3.5 mA rms (AC and DC) from any accessible conductive parts of the test set (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch or switches.

2.2.4 Water Resistance. The equipment shall not be required to be watertight, splashproof, or dripproof.

2.2.5 Dust Resistance. Dust resistance shall not be required.

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2.2.6 Salt Exposure. Salt exposure verification testing shall not be required.

2.2.7 Temperature Operating. The test set shall meet its performance and accuracy requirements in a minimum operating environment of -20°C to +55°C.

2.2.8 Relative Humidity. The equipment shall meet its performance and accuracy requirements while being operated in a non-condensing humidity range of ≤ 30% to ≥ 70%.

2.2.9 Random Vibration. The equipment shall conform to the random vibration requirements of MIL-PRF-28800 Class 3.

2.2.10 Transit Drop. The equipment shall conform to the transit drop requirements of MIL-PRF-28800 Class 3.

2.2.11 Warm Up Time. The test set warm up time shall be ≤ 30 minutes.

2.2.12 Environmental Changes. Changes in environmental conditions within the operating range of the test set will not require additional calibration, except for internal calibration procedures and signal path compensation, to meet performance requirements.

2.3 Electrical Power Sources. The test set shall operate from nominal commercial, military, and shipboard power sources as specified herein.

2.3.1 Single Phase, 120 Vrms. The test set shall operate at 120 Vrms, single phase AC (±10%) at line frequencies of 50 Hz, 60 Hz, and 400 Hz (±5%) in accordance with MIL-PRF-28800 Steady-State Conditions, Transient-State Conditions, and Interruption of Power Source requirements.

2.3.2 Single Phase, 240 Vrms. The test set shall operate at 240 Vrms AC (±10%) at line frequencies of 50 Hz and 60 Hz (±5%) in accordance with MIL-PRF-28800 Steady-State Conditions, Transient-State Conditions, and Interruption of Power Source requirements.

2.3.3 DC Internal Power Source. The test set shall operate from an internal DC power source in accordance with MIL-PRF-28800. The internal DC power source shall provide operating power when AC power is not available (i.e. flight line use). The internal DC power source may be either rechargeable or non-rechargeable and shall meet the specifications herein. Replacement of the DC internal power source shall not void or require calibration of the test set. The DC power source shall be commercially available.

2.3.3.1 Rechargeable Internal DC Power Source. The test set shall provide a means of charging the internal power source. The rechargeable DC internal power source shall provide a minimum of 8 hours of continuous operation of the unit within all specification requirements.

2.3.3.1.1 Recharge Time. The charger circuitry shall be such that a condition of charge which shall permit full continuous use of the unit, within all specification requirements, shall be reached in no more than twelve hours following a low DC source indication. Under no condition of discharge shall the charging rate applied to the rechargeable DC internal power source exceed the manufacturer's recommended maximum rate, nor shall an indefinite charging period result in any damage to the rechargeable DC internal power source or any other components.

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2.3.3.2 Non-Rechargeable Internal DC Power Source. A non-rechargeable DC internal power source shall be required to provide a minimum of 40 hours continuous operation of the unit within all specification requirements.

2.3.4 Power Consumption. Maximum power consumption of all components combined shall not exceed 40 watts.

2.4 Reliability. The design of the test set shall be such that under normal use and operation the test set does not fail within 4,000 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The test set shall have a calibration interval of at least 180 days.

2.6 Performance Characteristics. Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply with or without test leads, after a 30-minute warm-up period.

2.6.1 General Specifications. The test set shall provide a means to determine the serviceability of aircraft employing an AC capacitance fuel gauging systems. It shall consist of a single integral unit incorporating a transit case, direct current (DC) electrical resistance measuring circuitry, capacitance measuring circuitry, capacitance substitution circuitry (simulates tank unit and compensator capacitance), switches, wiring, cable assemblies, and accessories. The test set shall operate from an external AC power source and/or DC internal power source.

2.6.1.1 Test Set Functions. The test set shall be able to perform the following functions as described in subparagraphs 2.6.1.1.1 thru 2.6.1.1.4.

2.6.1.1.1 Capacitance Measurement. The test set shall provide AC capacitance measurements to determine the serviceability of aircraft capacitance type fuel quantity gauging systems. The test set shall automatically display the measured capacitance, (i.e. no necessary computation of the tank unit and the compensator). The capacitance measurement system shall have a range from 0 to 8000 pf; the zero measurement may be relaxed slightly but not more than 5 pf. The measured capacitance value shall have an accuracy within ±0.10% of the maximum value of each capacitance range. The values established up scale and down scale shall not deviate from that accuracy by more than 0.05%. The resolution shall be at least 0.025% of the range selected.

2.6.1.1.2 Simulated Tank Unit Capacitance. The test set shall provide functionality to simulate tank unit capacitance in the range from 0 to 8000 pf. The capacitance simulation shall have an accuracy within ±0.10% of the reading over 200 pf and ±0.2% below 200 pf.

2.6.1.1.3 Simulated Compensator Capacitance. The test set shall provide functionality to simulate compensator capacitance in the range from 0 to 390 pf. The capacitance simulation shall have an accuracy within ±0.10% of the reading over 200 pf and ±0.2% below 200 pf.

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2.6.1.1.4 Insulation and Tank Resistance Measurement. The test set shall provide functionality to measure the insulation and tank resistance. The resistance measurement shall have a minimum range from 0.1 Ω to 10,000 MΩ, ($1x10^{10}$ Ω). The resistance measured shall not deviate from the true resistance by more than 15%. The resolution shall be at least 0.1% of full scale. The DC voltage applied across the circuit shall not exceed 50 volts and the short circuit current, when the output receptacles are shorted together, shall not exceed 10 mA.

2.6.2 Test Measurement Sample Rate. The test set data acquisition measurement sample rate shall not be continuous but at intervals so as to minimize Re-Test OK, (RTOK) and Can Not Duplicate (CND) type of diagnostic errors.

2.7 Cables and Connector Types. The manufacturer shall supply test cables. Relocation of cables shall not be required when changing modes of operation (e.g. from capacitance to resistance). All cables for use with the GTF-6 and PSD60-1AF shall be completely compatible with the test set.

2.7.1 Tank Unit and Indicator Cables. All aircraft tank unit and indicator cables connectors shall be the BNC type. The cables shall be two impedance and length matched pairs; one pair with type BNC(m) connectors and the other pair with type BNC(f). The test cables shall be 10 feet in length.

2.7.2 Grounding. A means for grounding the test set to the aircraft shall be provided.

2.7.3 Power Cable. A power cable shall be provided for bench top use and/or charging of any internal DC power source while external power is utilized.

2.8 Indicators. All measurements shall be displayed on a 4.5-digit (minimum) lighted indicator with floating decimal point. The indicator shall be visible in bright sunlight and meet all requirements of Class 2 of the MIL-PRF-28800.

2.9 Manuals and Documentation. The equipment shall be delivered with operator manual, maintenance manual, calibration manual, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

2.10 Physical Characteristics. The size and weight of the test set shall be consistent with current commercial practices. The equipment shall be portable with the following physical characteristics.

2.10.1 Size. The test set shall be no larger than 18 inches long x 15 inches wide x 13 inches deep.

2.10.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a one-person lift of 37 pounds (or 16.8 kilograms).

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The test set provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 Warranty. The manufacturer shall offer this test set with a five (5) year warranty, as a minimum, on parts and labor to repair these instruments, should they experience a fault which is not obviously due to abuse or irresponsible actions. A label shall be affixed to the test set providing the shipping address for warranty service and the period of the warranty by date. A note shall be on the label stating, "Return shipping address must be enclosed."

4.4 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packaging, Labeling, And Marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094 or online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil.

6.1.2 The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

6.2 Ordering data.

a. CID document number and revision.

b. Product conformance provisions (see 4).

c. Packaging requirements (see 5).

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6.3 Keywords.

Class 2

Test Set

Type II

MILITARY INTERESTS:

Custodians:
Air Force – 99
Navy – SH

Civil agency:
GSA-FSS

Preparing activity:
Air Force – 84

Agent:
Air Force – 99

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